NeuroTraffic: Smart Traffic Control System

AI-Driven Adaptive Traffic Management

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March 22, 2026

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Outline i

1. Introduction

Introduction

- Cities are expanding rapidly, and road infrastructure is struggling to keep pace.
- Most traffic signals still operate on fixed timers, regardless of real-time traffic conditions.
- This project explores how AI can enable intersections to adapt dynamically.
- Smarter traffic control reduces commute time, fuel consumption, and emergency delays.

2. Problem Statement

Problem Statement

- Traditional traffic signals follow pre-set timing cycles.
- They cannot adapt to sudden traffic surges or empty lanes.
- This results in:
  - Unnecessary waiting at red lights
  - Heavy congestion during peak hours
  - Delayed emergency vehicle movement
- **Core Issue:** Lack of an intelligent system that dynamically adjusts signal timing based on live road conditions.

3. Objective

Objective

Main Goal:

- Develop an AI-powered adaptive traffic signal control system.

Specific Objectives:

- Real-time vehicle detection and tracking.
- Short-term congestion prediction using sequence models.
- Emergency vehicle prioritization.
- Illegal parking detection at intersections.

Target Outcome

Target Outcome:

- Reduce average waiting time by 30–35%.
- Maintain sub-second response latency.

4. Feasibility Study

Feasibility Study

Technical Feasibility

- Modern AI frameworks enable real-time inference.
- GPU acceleration ensures fast processing.

Economic Feasibility

- Utilizes existing CCTV infrastructure.
- Avoids expensive road-embedded sensors.

Operational Feasibility

- Intuitive dashboard for traffic monitoring.
- Clear visualization and alert system.

5. Requirements

Requirements

Functional Requirements

- Real-time vehicle counting (Cars, Buses, Trucks, Motorcycles).
- Lane-based detection using polygon masking.
- AI-based dynamic signal switching.
- Emergency vehicle green corridor feature.
- Illegal parking detection for stationary vehicles.

Non-Functional Requirements

- Low latency video streaming (¡ 1 sec delay).
- Scalable architecture for multiple intersections.
- Reliable logging and fault tolerance.

6. Design

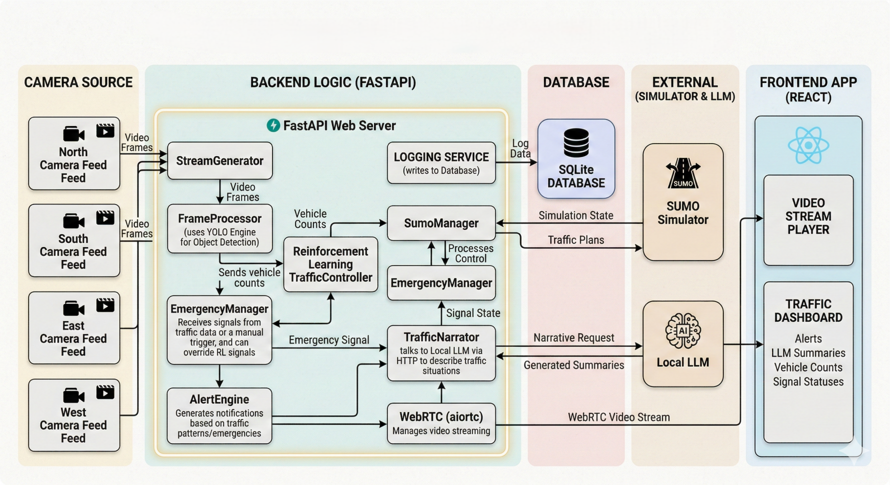
System Architecture (1/2)

1. **Perception Layer**
   - Processes camera feeds.
   - Applies lane masking.
   - Uses YOLO-based tracking for vehicle detection.
2. **Decision Layer**
   - Deep Q-Network for signal decision making.
   - Bi-LSTM model for congestion prediction.

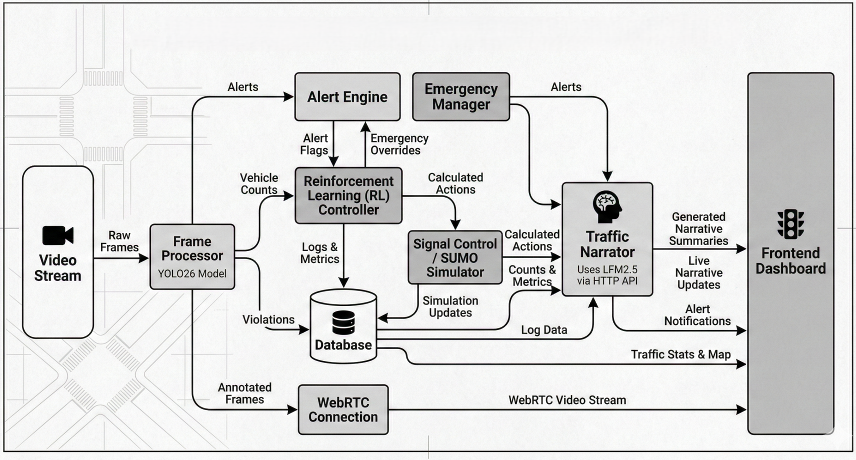
System Architecture (2/2)

3. **Communication Layer**
   - REST APIs for signal control.
   - WebRTC for live streaming.
4. **Presentation Layer**
   - Interactive React dashboard.
   - Real-time metrics and alerts.

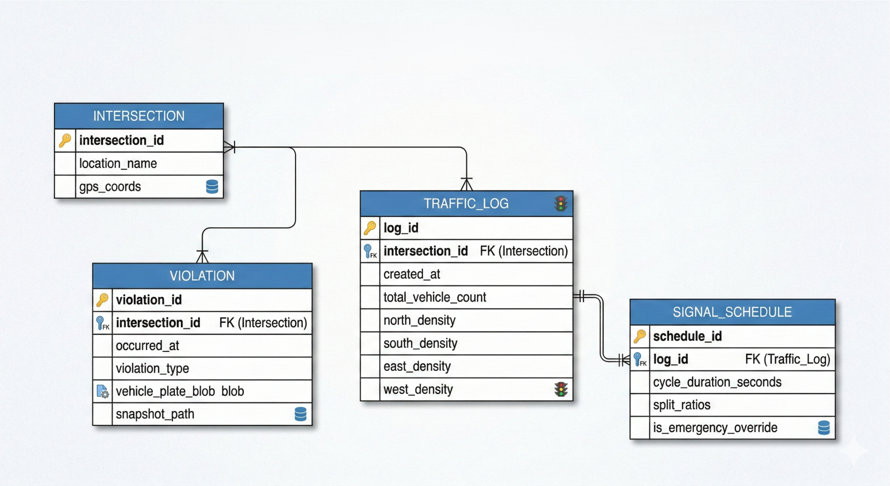
System Architecture Diagram



Data Flow Diagram



Entity Relationship Diagram



7. Implementation

Tools and Technologies

Backend

- Python (FastAPI, Uvicorn)
- PyTorch
- YOLO for vehicle detection
- OpenCV
- SUMO for simulation

Frontend

- React + TypeScript
- Tailwind CSS
- React Query
- Recharts for visualization

8. Testing and Results

Testing Approach

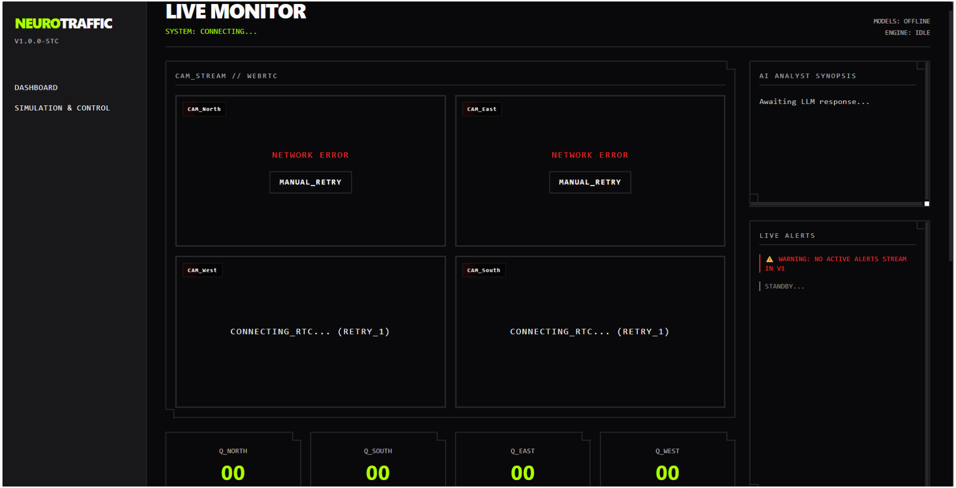
- Traffic simulation using SUMO.
- Comparison between static timer and AI controller.
- Evaluation over 1000 simulation steps.

Results (Simulation)

Metric	Static System	NeuroTraffic
Avg Queue Length	24 vehicles	$\sim$ 14 vehicles
Avg Wait Time	48 sec	$\sim$ 31 sec
Inference Latency	N/A	$<$ 40 ms

Observation: The AI-based controller significantly reduced congestion while maintaining low latency.

User Interface



9. Conclusion

Conclusion

- NeuroTraffic demonstrates how AI can make intersections smarter and adaptive.
- Simulation results show reduced waiting time and improved traffic flow.
- The system is scalable and suitable for smart city deployment.

Limitations

- Requires robust hardware for real-world deployment.
- Needs testing under varying weather conditions.

Future Work

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- Multi-intersection coordination.
- City-wide deployment model.
- Integration with smart city infrastructure.